EXPERIMENT-3

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1.COMMAND LINE INTERFACE:

PYTHON CODE:

```
import psutil


def show_battery():
    battery = psutil.sensors_battery()


    if battery is None:
        print("No battery detected")
        return


    print("\nBattery Percentage:", battery.percent, "%")


    if battery.power_plugged:
        print("Status: Charging")
    else:
        print("Status: Not Charging")


while True:
    print("\n--- Battery Manager CLI ---")
    print("1. Show Battery Status")
    print("2. Exit")


    choice = input("Enter choice: ")


    if choice == "1":
        show_battery()
```

```
    elif choice == "2":
        break

    else:
        print("Invalid choice")
```

OUTPUT:

```
PS C:\Users\Jai varadhan\OneDrive\Desktop\UID-BATTERY> python command_line.py

--- Battery Manager CLI ---
1. Show Battery Status
2. Exit
Enter choice: 1

Battery Percentage: 70 %
Status: Not Charging

--- Battery Manager CLI ---
1. Show Battery Status
2. Exit
Enter choice: 2
PS C:\Users\Jai varadhan\OneDrive\Desktop\UID-BATTERY>
```

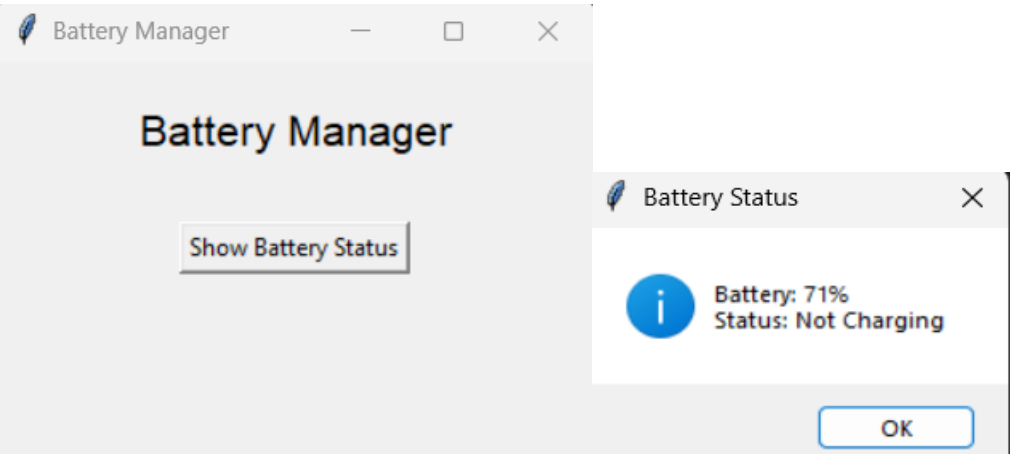
2. GRAPHICAL USER INTERFACE:

PYTHON CODE:

```
import psutil

import tkinter as tk

from tkinter import messagebox


def show_battery():

    battery = psutil.sensors_battery()


    if battery is None:

        messagebox.showinfo("Battery", "No battery detected")
```

```
        return

    percent = battery.percent
    status = "Charging" if battery.power_plugged else "Not Charging"

    messagebox.showinfo(
        "Battery Status",
        f"Battery: {percent}%\nStatus: {status}"
    )

root = tk.Tk()
root.title("Battery Manager")
root.geometry("300x200")

tk.Label(root, text="Battery Manager", font=("Arial", 16)).pack(pady=20)

tk.Button(root, text="Show Battery Status", command=show_battery).pack(pady=10)

root.mainloop()
```

OUTPUT:



3.VOICE USER INTERFACE:

PYTHONE CODE:

```
import psutil
import speech_recognition as sr

def get_battery():
    battery = psutil.sensors_battery()

    if battery is None:
        return "No battery detected"

    percent = battery.percent
    status = "Charging" if battery.power_plugged else "Not Charging"

    return f"Battery is {percent} percent and {status}"

recognizer = sr.Recognizer()

print(" 🎤 Say 'battery status' or say 'exit'")

while True:
    try:
        with sr.Microphone() as source:
            print("\nListening...")
            recognizer.adjust_for_ambient_noise(source, duration=1)
            audio = recognizer.listen(source)

        text = recognizer.recognize_google(audio).lower()
        print("You said:", text)

        if "battery" in text or "status" in text:
```

```
        print(get_battery())

    elif "exit" in text:
        print("Exiting program...")
        break

    else:
        print("Command not recognized")

except sr.UnknownValueError:
    print("❌ Could not understand audio")

except sr.RequestError:
    print("❌ Internet / Speech service error")

except Exception as e:
    print("Error:", e)
```

OUTPUT:

```
PS C:\Users\Jai varadhan\OneDrive\Desktop\UID-BATTERY> python voice_interface.py
🎤 Say 'battery status' or say 'exit'

Listening...
You said: battery status
Battery is 69 percent and Not Charging

Listening...
You said: exit
Exiting program...
PS C:\Users\Jai varadhan\OneDrive\Desktop\UID-BATTERY>
```